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Impact of viewer engagement on gift-giving in live video streaming

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ABSTRACT

This study investigates the effect of viewer engagement on gifting items to a streamer in a live video streaming. Data were collected from AfreecaTV, a leading live video streaming platform in South Korea, to examine viewer engagement and gift-giving behavior. After analyzing 2,294,837 viewers over a three-month period, the empirical results provide evidence that viewer engagement is positively associated with gift-giving decisions. However, the impact of viewer engagement on the amount of gifts purchased is different depending on how the engagement is measured (i.e., by stream or by channel). This study empirically proves that the motive for socialization has a high correlation with gift-giving behavior, which is considered as commoditization of a viewer's social interaction while consuming media. The study concludes with a discussion on practical implications for live video streaming services and suggestions for future research.

1. Introduction

Live video streaming services[a] such as Twitch, YouTube Live, Facebook Live, AfreecaTV, YY.com, and Douyu TV have created an environment where anyone can broadcast live content. Every user can create her/his own content or participate in other viewers' live video streams. In addition to chatting with other viewers, viewers of live video can also communicate with the broadcaster (or streamer) through various ways: by chatting, by gaming as a team, or by sending gifts.

Gift-giving is a unique business model developed in the context of live video streaming. In real-time (i.e., during a broadcast), viewers can gift streamers any item or any amount of virtual money, some of which can be exchanged with real currency. Gift-giving is a voluntary action since viewers can watch almost all the streams without giving a gift. In return, the giver receives the streamer's gratitude, which the latter highlights over a public chat, as a non-monetary reward. Therefore, gift-giving is an opportunity for viewers to attract everyone's attention. Moreover, for those who are part of a community that shares experiences about the streamer, sending gifts enables them to have a feeling of superior social status over other ordinary viewers, as they can communicate with the streamer (Hamilton et al., 2014; Oh and Choi, 2017).

Several live video streaming services offer a gift-giving feature. Viewers can buy items or virtual money for giving gifts. Twitch offers "Bit," YouTube Live offers "Super Chat," AfreecaTV offers "Star Balloon," and YY.com offers "Rubi." YY.com earned about $230 million in the last quarter of 2015 from gifts (Xiang, 2016) and AfreecaTV doubled its revenue from gifts in 2016 from 2014 (AfreecaTV, 2017). Twitch, which adopted the gifting function in 2016, sold $8 million worth of "Bits" in the first six months of their introduction (Twitch, 2017c). Since everyone can support any creator, gift-giving is also an opportunity for small channels that

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[a] The term, live video streaming (Edge, 2013; Hu et al., 2017) is also called live video broadcasting (Yang, 2016), social video streaming (Twitch, 2017a), or just live streaming (Hamilton et al., 2014; Lessel et al., 2017). In this study, we use the term "live video streaming."

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cannot find an official sponsorship (Perez, 2017) or other revenue sources. Thus, the gift-giving business model becomes important for live video streaming platforms as a highly profitable growth opportunity.

Despite its importance, gift-giving has not been explored sufficiently in research studies on live video streaming. Only a few studies have investigated motivational and behavioral factors that encourage viewers to purchases for streamers (Gros et al., 2017; Sjöblom and Hamari, 2016). Although Kim et al. (2016) examined the intention to purchase gifts and Zhu et al. (2017) described behavioral patterns of gift senders in broadcasting, none of the previous studies explored the quantified consequences of social interactions among streamers and viewers.

This study examines gift-giving behaviors in live video streaming through the lens of viewer engagement in the Korean platform, AfreecaTV. As the top live video streaming platform in South Korea, AfreecaTV has been operating a gift-giving business model since 2006, with 70% of its revenue coming from it (AfreecaTV, 2017). After analyzing the individual-level field data of 2,294,837 viewers, this paper suggests several implications for live video streaming service operators and streamers. The results emphasize that the importance of social engagement through gift-giving and the interactive culture of live video streaming may be a competitive advantage for live video streaming services over conventional media.

The rest of paper is organized as follows. Section 2 is a literature review of engagement and gift-giving behavior. Section 3 presents the hypotheses and the research model. The description of the data and the methodologies are shown in Section 4. Section 5 presents the results of the analysis, and a discussion. Section 6 provides the conclusion and implications.

2. Literature reviews

2.1. Live video streaming

Live video streaming services have grown tremendously around the globe. According to a 2017 Magid report, 48% of Internet users in the US watch live videos at least once a week, and a quarter of them consume live videos at least once a day (Magid, 2017). Twitch is one of the pioneers of live video streaming, focusing on video games and e-sports streaming, with about 1 million streamers interact with more than 100 million viewers every month (Twitch, 2017b). In China, which has the largest live video streaming market, 324.8 million Internet users (or 45% of its online users) access live streaming content, more than 200 live video streaming apps are available, and the live video streaming market grew at 180% in 2016 (Chadha, 2016; Si, 2017; Xiang, 2017).

Live video streaming enables synchronous communication between a streamer and viewers, and between a viewer and other viewers (Holt et al., 2016). For example, when a streamer asks viewers which game they want to watch, viewers offer suggestions via live chat. Viewers also talk to each other about their experiences of the streamer's previous broadcast. Sometimes, the streamer invites viewers to play a game as a team. These new participatory features assist viewers to become involved in the streamer's content, even as creators (Hamilton et al., 2014; Lessel et al., 2017; Oh and Choi, 2017).

Early studies on live video streaming focused on e-sports viewership and found underlying social activities based on online communities (Cheung and Huang, 2011; Edge, 2013; Scholz, 2011). Following studies noted diverse ways of participation centered on a streamer. Streamers try to interact with their audiences, and viewers who watch the same streamer's broadcasts share their experiences and build a social relationship (Hamilton et al., 2014; Smith et al., 2013). Recent studies showed insights into viewers' behaviors in the context of their social motivations (Hu et al., 2017; Sjöblom et al., 2017).

However, there are a small number of researches dealing with purchase and social behavior in live video streaming. Sjöblom and Hamari (2016) found that the sense of being connected is the only motivational aspect that compels viewers to pay for a subscription. Gros et al. (2017) indicated that socialization is more significant than information and entertainment factors, for paid subscription and donation. Kim et al. (2016) showed that a friendly feeling towards a streamer increases the viewer's intention to purchase gifts through a sense of satisfaction. Zhu et al. (2017) identified some patterns of giving gifts during live video streaming: only a small number of viewers send a majority of the gifts and viewers are motivated to send gifts after observing other viewers' gift-giving behaviors. Even though previous researches provided insights about participatory culture and the social-psychological approach, they rarely reflected on individuals' interactions in live video streaming and the subsequent outcomes.

2.2. Viewer engagement

Compared to traditional media, the key differences in a live video streaming service are: it allows viewers to interactively engage in content (i.e., the streamers' content) and with other viewers in real time to a considerable extent. These arise from the main feature of the service that live broadcasting and participatory culture (i.e., social interaction) exist at an unprecedented level (Witkowski et al., 2016). Engagement is defined as the user's investment of physical and psychological energy to fulfill certain psychological needs such as information seeking, socialization or social interaction, and entertainment (Astin, 1984; Ruggiero, 2000). The engagement contains all sorts of user attention and involvement with media, and makes available viewers to consume, produce, or disseminate information (Deuze, 2003; Hargittai and Walejko, 2008; Napoli, 2011). For instance, viewers in Twitch, AfreecaTV, Douyu TV, and other live video streaming services interact with streamers who create content and with anonymous people who are simultaneously watching the broadcast, via live chat. This type of engagement enables content to be created in a real-time basis and distributed widely. In particular, AfreecaTV enables viewers to not only participate in live chats with others, but also recommends content that viewers have watched. This reflects streamer rankings and therefore, influences on other viewers' selection of content. Since viewer recommendation is reflected in their rankings, whenever viewers recommend the contents, streamers check their user IDs, react and thank them by calling out their IDs, dancing wackily, or by doing other gestures. This type of viewer engagement is

different from that in live chats as it is in the form of real-time interactions between the viewers and a streamer. Computer-mediated communications, which encourage viewers to engage, are appearing in various forms in live video streaming services.

Viewer interaction exists while watching television, and related research has been conducted as well. Jensen and Toscan (1999) defined the concept of interactive television (ITV) as a "two-way television in which the viewer can make programming choices and produce user input" and it has been commonly leveraged in subsequent studies. Based on this definition of ITV, van Dijk et al. (2003) stipulated the level of interactivity concerning television watching in four steps and presented the specific activities viewer performed in each step. Andrejevic (2008) examined viewer participation as a perspective of "value-enhancing labor" for television producers. Although real-time interaction may not have occurred in the traditional television context, there existed two-fold interactions: fans took on part of the work of making a show interesting for themselves and provided "instant" feedback to producers. However, these interactions are much closer to participation as providing free labor rather than the engagement. While real-time user engagement helps to produce and propagate content seamlessly in live video streaming services, traditional TV has fundamental differences, such as non-real-time interaction, production by direct participation rather than engagement, and non-spreadability, even though the research applied the term "instant" feedback. Therefore, the traditional approach of media researches has limitations in explaining today's phenomena. According to the interactivity framework of van Dijk et al. (2003), the interaction investigated in Andrejevic (2008) belongs to level 3, in accordance with the forms of reaction/commentary to, participation in, and contribution to programs. However, the interaction in live video streaming corresponds to level 4 in terms of communication to television programs (creators), viewers, or user communities. That is to say, while viewer engagement is just reaction or commentary to programs after watching them in traditional media, users of live video streaming communicate with each other in real time and thus may create programs together by participation.

In addition, Ksiazek et al. (2014) studied the interactivity between online news videos and users, reflecting the rapidly developing online media consumption context. They defined engagement as users commenting on or displaying conversational behaviors with others while watching YouTube news videos, and examined how popularity, news type, and objectivity aspects affect these engagement behaviors. There have also been viewer interaction studies in live video streaming services. Hamilton et al. (2014) explored the motivations of viewer interactions with others.

Although the existing studies are academically and practically meaningful in explaining interactivity in media consumption, there are some limitations in explaining viewer engagement in live video streaming. The reason for these limitations is that they have not investigated factors actually observed in live video streaming services. This has provided irrelevant implications to practitioners. More importantly, while there have been attempts to find influential factors for engagement such as motivation, there has been no attempt to improve understanding of the consequences of viewer engagement, instead of looking for the antecedents. The thing that we need to focus on is that live video streaming services enable viewers' attention and social interactions to be "commodified" (Witkowski et al., 2016). It appears as a form of viewers' gift-giving behavior called "donation" in Twitch. The viewers can purchase a gift-item on the service platform and send it to the streamer they are watching. It gives the streamer a monetary or non-monetary benefit depending on the types of items purchased. Since gift-giving behavior is on the premise of the desired or established social relationships between the receiver and giver (Belk, 1979), it seems to be related to viewer engagement that exists due to the new communication methods in live video streaming. The main reason for viewer engagement in media consumption is the need for social interaction (Chung and Yoo, 2008; Yoo, 2011). Gift-giving "commodifies" the need for social interaction and makes it appears as purchasing and sending items. Consequently, it is expected that there is a strong correlation between viewer engagement and gift-giving behaviors. However, research to identify this phenomenon has not been carried out. In particular, gift item sales are the additional source of revenue, besides advertising revenue, only for live video streaming providers. Therefore, identifying the relationship between engagement and gift-giving has considerable importance for such businesses. This study is the first approach in engagement research stream in that it identifies the relationship between viewer engagement and gift-giving which is the commodified social interaction.

2.3. Gift-giving behavior

Gift-giving is a voluntary investment of time and money to help someone, and exists in both non-monetary and monetary forms (Harbaugh, 1998; Prendergast and Stole, 2001). The fundamental motivation for gift-giving is derived from "status-seeking" behaviors. Although there is no direct benefit to the giver, the motivation to enhance her/his own status leads to a social passion to invest her/his time and effort for others (Harbaugh, 1998; Sherry, 1983). The status-seeking behavior may be explained as either enjoying an economic and social advantage or as obtaining psychological and emotional satisfaction (Perretti and Negro, 2006). These two reasons are not perfectly mutually exclusive, and thus, act mostly at the same time. But the economic and social reason is to strive for a higher status (Lin, 2002), and the emotional reason is for egotistic rewards or happiness (Dunn et al., 2008; Emerson, 1962). Whatever the reason is, most gift-giving appears as a form of socialization, based on motivation from status-seeking behaviors.

The gift-giving behavior happens in online communities for the same reason. The reasons for providing non-monetary gifts such as opinion, information, and advice are helpful in seeking a higher status such as prestige, honor, and deference (Lampel and Bhalla, 2007). Goode et al. (2014) explored the gift economy phenomenon from the perspective of users' social statuses on Massively Multiplayer Online Games (MMOGs), and found a high correlation between the giver's social status enhancements in the future and her/his gift-giving behavior.

Even at a primitive level of research, there are studies on the gift-giving phenomena in live video streaming services as well (Gros et al., 2017; Zhu et al., 2017). Zhu et al. (2017) identified the phenomena on the Chinese platform Douyu TV as an exploratory study, and presented descriptive statistics of viewers' behaviors. Gros et al. (2017) examined the gift-giving (donation) behavior of Twitch

users in a more specific manner. They classified motives to use Twitch, into entertainment, information, and socialization, and investigated the impact of these motives on average usage time and money spent, through a survey method. The time and money spent (i.e., on gift-giving) were highly correlated, and the socialization motivation had a significant effect on spending. The study is meaningful as it showed the relation between gift-giving and socialization in live video streaming services. In particular, gift-giving can be seen as a commoditization of viewers' social interactions, as Witkowski et al. (2016) asserted, which was a significant attempt to discover its relevance to socialization. Yet, their study has limitations because of the survey method used. They did not explore the relationship between engagement and gift-giving behaviors actually observed in the platform. During the research process, they measured the amount of money respondents spent in the past and their motivation, by asking why they used live video streaming services, regardless of time. Therefore, the causal relationships they found were problematic. This creates a limitation in providing accurate and concrete explanations about gift-giving behaviors and presenting meaningful implications. In addition, the overall accuracy of the research is not ensured as the questionnaires were arbitrarily organized rather than being based on existing literature. Thus, we strongly consider the necessity to improve these limitations and uncover the actual relationship between engagement and gift-giving phenomena using actual log data.

Based on the theoretical background, this study poses specific research questions as followings: Does viewer engagement affect gift-giving behavior? Does the effect vary by the level of engagement? In order to answer these questions, this study established a research model and developed hypotheses in the following section.

3. Hypothesis development

In the online business context, the concept of engagement assists service providers to estimate users' reactions to their services such as satisfaction and intention to purchase (Alnawas and Aburub, 2016; Hutter et al., 2013). Brettel et al. (2015) identified that users' engagement on a company's Facebook page raises both short-term and long-term sales. Kilger and Romer (2007) showed that the social feature of engagement on the Internet is positively correlated with the likelihood of product purchase. Oestreicher-Singer and Zalmanson (2013) note that active users of an online community are more likely to join paid subscription than users who consume content only. This is because users' payment and willingness to pay could lead to their participation (engagement) in community or social behaviors in the process of consuming content, not the content consumption itself. The engagement may yield loyalty (attitude towards brand, community, service or platform), which make users to pay (Dick and Basu 1994). It makes people consciously pay money and makes them insensitive to price (Raju et al. 1990; Zeithaml et al. 1996). The evidence of positive effects of engagement on purchase decisions has been drawn. Therefore, our first hypothesis is:

Hypothesis 1. *Viewer engagement positively affects gift-giving behaviors.*

Only a few paying customers are highly valuable for businesses. Therefore, what makes viewers purchase more is a different problem from what leads viewers to decide payment (Voigt and Hinz, 2015). Similarly, in the context of live video streaming services, 0.5% of gift senders generate about half of the revenue from gifts and most senders give the least valuable gift (Zhu et al., 2017). Besides, gifts act as an economic signal, which is invested for a desirable relationship (Camerer, 1988). Social relationships require givers to choose appropriate gifts regarding events, seasons, and prices (Belk, 1979; Wolfinbarger, 1990). Therefore, among viewers of the paid group, a positive association between viewer engagement and the amount of purchase is expected, because the engagement is based on the degree of social relationship (Yoo, 2011). Therefore, our second hypothesis is:

Hypothesis 2. *Viewer engagement positively affects the amount of gift-giving.*

4. Methodology

4.1. Data collection

The dataset is randomly sampled from a group of viewers who watched live streams for more than 20s in AfreecaTV between October 1 and December 31, 2016. The company name, AfreecaTV, means "a(nybody) free (broad)casting TV." As the top live video streaming platform in South Korea, its mobile application has been downloaded 53 million times, and 6 million unique viewers visit AfreecaTV every month. While most of the existing media revenues come from advertising, more than 70% of AfreecaTV's total earnings is from gift items. In 2006, the company launched its platform with features where viewers can send gifts to streamers. It earned almost $50 million from gift items in 2016, which has doubled since 2014 and tripled since 2013. The profit of the company's top streamers also grew about 40% annually (AfreecaTV, 2017). As the gift-giving business model has been well implemented by AfreecaTV, it is an optimal platform to examine the relationship between viewer engagement and gift-giving behaviors.

After removing users who signed up during the period of observation between October 1 and December 31, 2016 and are aged less than 10 or more than 70, due to the validity of the customer profiles, a total of 2,294,837 viewers remain in the data set. Viewers in their 20s form the largest age group (44%), and male viewers are more than three-fourths (over 77%) of the total users. The share of mobile-only viewers (62%) is ten times larger than PC-only viewers (6%). A viewer's "main genre" is the most watched genre from her/his records. Games are preferred by almost half of the viewers (49%), followed by talks (37%), and others. One-fifth (21%) of the viewers joined AfreecaTV less than a year ago, and one-third (34%) has stayed with the platform for more than five years. For the three-month period of focus, the first quintile of viewing time is 100 h. Table 1 describes the details of the sample.

Table 1
Sample description.

Variable		Percent
Age	10s	12.18
	20s	43.92
	30s	20.66
	Over 40s	23.24
Gender	Male	77.45
	Female	22.55
Channel	PC only	6.02
	Mobile only	62.25
	PC/Mobile both	31.73
Days since a viewer joined	Less than a year	21.29
	1–5 years	43.44
	5–10 years	17.49
	More than 10 years	17.78
Viewing time	< 10 h	39.68
	10–100 h	39.50
	More than 100 h	20.82
Main genre	1. Game	48.94
	2. Talks	37.33
	3. Eating/Cooking	5.05
	4. Life	3.03
	5. Mobile game	2.19
	6. Sports	0.95
	7. Music/K-pop	0.71
	8. Animation	0.49
	9. Education	0.33
	10. News/Current affairs	0.3
	11. TV	0.26
	12. Finance	0.26
	13. Beauty/Fashion	0.09
	14. Others	0.07
Total	2,294,837	100

4.2. Measurement

In AfreecaTV, viewers can purchase three kinds of items: “Star Balloon,” “Gold,” and “Quick View.” A viewer may present a streamer with a “Star Balloon,” which includes monetary rewards. “Gold” is a non-monetary gift that can be used for buying functional items, such as increasing audience capacity or exposing a broadcast on the top of the list. It cannot be converted into money. “Quick View” enhances the viewer’s convenience of using the platform, such as, by removing ads. Since the revenue of “Star Balloon” is 30 times bigger and the ratio of “Star Balloon” users is two times more than those of other items (AfreecaTV, 2017), this study focuses on “Star Balloon.”

The dependent variables are whether a viewer purchased a “Star Balloon” and the amount of “Star Balloons” purchased during the three-month period. Although these variables indicate the act of purchasing gifts and not gift-giving, the items may be presented only after their purchase and cannot be exchanged with other items of AfreecaTV. Hence, it is reasonable to assume that all viewers buy “Star Balloons” to present them to streamers. 10 “Star Balloons” cost about $1, and viewers can send any amount they want. When a viewer sends “Star Balloons” to a streamer, a message, that includes the giver’s nickname and the volume, pops up in the chat windows with a graphic, depending on the number of “Star Balloons.”

Viewer engagement is measured through chat and recommendations. Viewers may not only simultaneously create content via live chat with a streamer and other users, but may also recommend a live video once in a day for each broadcast in AfreecaTV. In live video streaming services, one streamer (i.e., a single ID) has one channel, and there are multiple streams (i.e., broadcasts) in one channel (see Fig. 1). Viewers may watch multiple streams belonging to various channels or merely the streams in a certain channel. Therefore, it is necessary to distinguish between content engagement and channel engagement. The number of streams that engage in activities such as chatting or recommendations can be described as content engagement. The number of channels participating in chatting or referrals can be described as channel engagement. Whereas content engagement is a concept similar to viewer engagement, in existing media consumption (Ksiazek et al., 2014), channel engagement generally corresponds to the concept of brand engagement (Hollebeek, 2011). Thus, the smaller the number of engaged channels, the higher the commitment or interactivity to a particular channel. For example, if a viewer chatted (i.e., engaged in a stream via chat) in 10 streams by one streamer for the three months, the viewer’s engagement in streams is measured as 10 and the engagement in channels is measured as one. Therefore, this study suggests four measurements of engagement to reflect both content and channel engagement as followings: *1) The number of streams* (i.e., *broadcasts) that a viewer chatted during the period* (i.e., *engaged in a stream via chat); 2) The number of streams that a viewer*

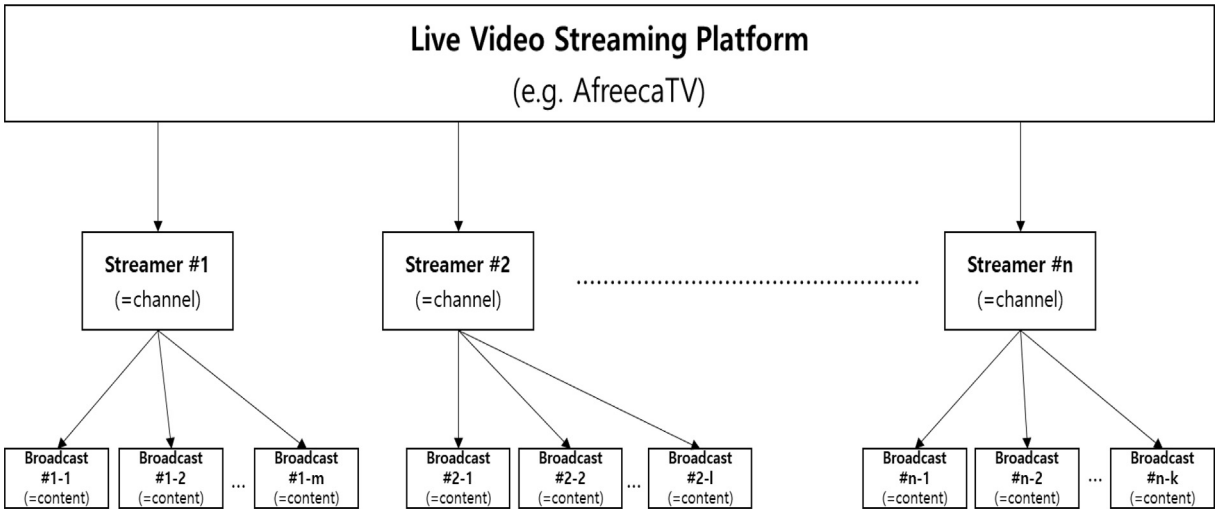


Fig. 1. Structure of live video streaming platform.

*recommended during the period (*i.e., *engaged in a stream by recommending it); 3) The number of streamers (*i.e., *channels) that a viewer chatted in/with during the period (*i.e., *engaged in/with a channel via chat);* and *4) The number of streamers that a viewer recommended during the period (*i.e., *engaged in a channel by recommending it).*

The control variables are viewer information (e.g., age, gender), days since a viewer joined, viewing time, previous experience on item purchase, and preference on genre. Variables of experience on item purchase are binary and represent whether a viewer purchased an item (e.g., Star Balloon, Gold, or Quick View) or not, in the three months before the period of focus. In our dataset, the portions of "Gold" and "Quick View" are much smaller than those of "Star Balloon," but it is expected that previous experience on item purchase may affect viewers' gift purchases, and thus, we set up the experience of each item purchase as a control variable. The details of the variables are provided in Table 2.

5. Data analysis and results

During the period of observation, 9.8% of the viewers had purchased gifts, and the average purchase amount was 144.51 items, which cost $13 roughly. Binary logistic regression is used for the dichotomous dependent variable of purchase, and negative binomial regression is applied to the purchase volume variable, which is highly skewed with many zeros (Greene, 2011). To reduce the skewness of the variables, viewing time by seconds is log-transformed (i.e., log_sec), and four engagement variables (i.e., rt_chat_stream, rt_recommend_stream, rt_chat_channel, and rt_recommend_channel) are square root-transformed due to their zeros. The variable of days since a viewer joined is added with its squared term (i.e., join_days and join_days_sq) to observe a potential quadratic effect. 11.7% viewers purchased items during the three months prior to the focal period. The descriptive statistics are shown in Table 3.

Table 4 provides the results of regression analysis. Content engagements through both chat and recommendations were positively associated with the purchase of gift items ($\beta = 0.119$, $p < 0.001$ for chat; $\beta = 0.103$, $p < 0.001$ for recommendations), and channel engagements through both chat and recommendations were positively associated with the purchase of gift items ($\beta = 0.162$, $p < 0.001$ for chat; $\beta = 0.168$, $p < 0.001$ for recommendations). Thus, H1 was fully supported. In Model 2-B, channel engagement through recommendations positively affected the amount of gift purchases as we expected ($\beta = 0.0812$, $p < 0.001$), whereas we found that channel engagement through chat was negatively related to the amount of gift purchases ($\beta = -0.0868$, $p < 0.001$). Content engagement through chat and recommendations has a positive effect on the amount of gift purchases in Model 2-A ($\beta = 0.0147$, $p < 0.001$ for chat; $\beta = 0.0508$, $p < 0.001$ for recommendations). Due to these inconsistent results, H2 was partially supported. We also found that the amount of gift purchases can vary depending on the engagement type and in such a case, the results of channel engagement through recommendations has an opposite direction to that of channel engagement through chat.

Viewers in their 30s have the greatest impact on both purchase and the quantity of purchase, when the age variable is inserted as a categorical variable as teenagers (baseline), 20s, 30s, and above 40s. Men are less likely to purchase than women ($\beta = -0.304$ in Model 1-A, $\beta = -0.245$ in Model 1-B, and $p < 0.001$), but the amount purchased was larger than that by women ($\beta = 0.109$ in Model 2-A, $\beta = 0.102$ in Model 2-B, and $p < 0.001$). When viewers watched content for a longer duration, both the probability of purchase and the amount of gifts purchased increased ($\beta = 0.155$ in Model 1-A, $\beta = 0.0688$ in Model 1-B, $\beta = 0.302$ in Model 2-A, $\beta = 0.156$ in Model 2-B, and $p < 0.001$). Among the three variables for item purchase experiences, "Star Balloon" and "Gold" were positively influential to gift purchase and the amount of gifts purchased. However, "Quick View" had a positive effect on gift purchase ($\beta = 0.0841$ in Model 1-A, $\beta = 0.0805$ in Model 1-B, and $p < 0.001$), whereas it had no significant effect on the amount of gifts purchased. We examined the relative effect of each genre on the dependent variables by defining game as a baseline. Viewers who like talks, eating/cooking, and music/K-pop are more likely to purchase gifts. However, viewers who prefer sports, news, finance, and

Table 2
Variable description.

Variable		Description
Gift purchase	Purchase or not	Whether a viewer purchased "Star Balloon" during the period
	The amount of purchase	The number of "Star Balloons" purchased during the period
Engagement	Engagement in stream: Chat	The number of streams in which a viewer chatted during the period
	Engagement in stream: Recommendation	The number of streams which a viewer recommended during the period
	Engagement in channel: Chat	The number of streamers with which a viewer chatted during the period
	Engagement in channel: Recommendation	The number of streamers which a viewer recommended during the period
Viewer information	Age	From 10 to 70, and other samples are removed due to validity of profile
	Gender	Male = 1, Female = 0
	Mobile only	1: Users who watch AfreecaTV via mobile only, 0: others
	Days since a viewer joined	The number of days since a viewer joined
Viewing time	Viewing time (second)	The number of seconds during the period
Previous experience on purchase	Star Balloon	This is a binary variable showing whether a viewer purchased "Star Balloon" during the last three months before the period. "Star Balloon" can be gifted in any amount, and can be exchanged for cash
	Gold	This is a binary variable showing whether a viewer purchased "Gold" during the last three months before the period. "Gold" is virtual money to buy "Sticker" items, serving live video streams like promotions on the main page
	Quick View	This is a binary variable showing whether a viewer purchased "Quick View" during the last three months before the period. "Quick View" helps viewers to navigate streams easily by offering features such as ad removal during the subscription
Preference on genre	1. Game	Streamers play games alone or with viewers or other streamers; Aion, Blade & Soul, Black Squad, Bubble Fighter, Cart Rider, Cyphers, Diablo3, Dungeon & Fighter, FIFA Online 3, GTA5, Hearthstone, Heroes of the Storm, League of Legends, Lineage, Mabinogi, Maple Story, Minecraft, Overwatch, Special Force, StarCraft, StarCraft 2, Sudden Attack, Tales Runner, Tekken, Tera, Tree of Savior, World of Tanks, World of Warcraft, Steam games, classic games, fighting games, sports games, video games, and other games
	2. Talks	Streamers talk with viewers about any topics, broadcast their life activities, or show performances
	3. Eating/cooking	Streamers eat or cook food
	4. Life	Hobby, health, and religion
	5. Mobile game	Clash of Clans, Pokémon GO, Clash Royal, Cookie Run, Seven Knights, Raven, Monster Taming, Everybody's Marble, and others mobile games
	6. Sports	Streamers perform as commentators; soccer, basketball, volleyball, baseball, martial arts, and others
	7. Music/K-pop	Music and K-pop streaming
	8. Animation	Streamers review animations, dub themselves, or just stream animations; Dubbing and review on animation, K, Naruto, Hunter X Hunter, A Certain Magical Index, World Break: Aria of Curse for a Holy Swordsman, Chuuka Ichiban, Space Dandy, and other animations
	9. Education	Streamers teach or share their live self-study sessions; language, self-study, and hobby
	10. News/Current affairs	Streamers share their opinions about current affairs
	11. TV	TV contents streaming
	12. Finance	Streamers share their portfolio and broadcast what they trade
	13. Beauty/Fashion	Streamers share their tips about beauty and fashion
	14. Others	Business, streamer award 2016, shopping, and others

life genres are less likely to purchase gifts.

6. Discussion

The hypotheses are consistent with what we expected except for the relationship between channel engagement through chat and the amount of gifts purchased, which has a negative effect. Since every viewer watched at least one broadcast during the period we sampled, this negative coefficient implies that fewer the streamers (i.e., channels) a viewer chatted with, more the gifts purchased. This phenomenon can be interpreted as a kind of fandom. Hu et al. (2017) mentioned that the link between a streamer and viewers is similar to a "celebrity-fan relationship" (p. 603). A fan community built around a streamer requires members to show their loyalty or to identify themselves among other viewers in various ways, such as giving gifts to the streamer or including the streamer's symbol or nickname in their username. Therefore, viewers who present gifts more than general users are likely to be fans. They are not willing to chat in other streamers' broadcasts, and focus on a few channels which they are fans of. The nature of giving gifts explains this phenomenon as well. A giver's anticipated or established social relationship with the receiver, decides what an appropriate gift may

Table 3
Descriptive statistics.

Variable	Mean	Std. Dev.	Min	Max
star_purchase_ox	0.098	0.297	0	1
star_purchase	144.51	4,343.87	0	2,875,500
age	31.28	11.753	10	70
join_days	1,752.598	1,578.575	0	5524
join_days_sq	5,563,495	7,907,229	0	30,500,000
log_sec	10.890	2.164	2.996	17.441
rt_chat_channel	2.141	2.183	0	44.978
rt_recommendation_channel	2.071	1.937	0	50.329
rt_chat_stream	3.204	3.922	0	73.621
rt_recommendation_stream	3.050	3.422	0	131.754
ox_star_3	0.096	0.295	0	1
ox_gold_3	0.007	0.084	0	1
ox_quick_3	0.034	0.181	0	1

be and how much effort should be invested (Camerer, 1988; Sherry, 1983). Thus, the intensity of the relationship between a giver and a streamer can affect the amount of gifts. Viewers who choose, and concentrate on establishing rapport with one or few streamers, tend to buy and give more gifts than viewers who chat around with many streamers. The reason why "chat" is more focused on several channels than "recommendation" is that it is necessary to spend more time in "chat" than "recommendation," and therefore, fans are willing to spend more of their time and energy to chat with a streamer and others.

There was a difference in the gift purchase behavior and the amount of gifts purchased according to gender. By observing the results in detail, men displayed lesser purchasing behavior than women. Women had a higher tendency to purchase items than men However, their purchase amounts were higher than those of women. This is primarily because gender difference plays an important role in hedonic consumption, and women respond more promptly to sensual cues that exist in the buying environment (Tifferet and Herstein, 2012). In AfreecaTV, there are various sensual cues during the broadcasts, and therefore, female users respond to the cues often and buy "Star Balloons" more frequently. Previous literature has documented that women have a greater level of risk-aversion (Dwyer et al., 2002). Therefore, their purchase amounts may be smaller.

This study investigated the relative effects of genres based on the game genre, which is a representative genre of live video streaming services (i.e., it has the highest percentage of viewership at 48.94%). In addition, there were significant differences based on genres. First, talks, eating/cooking, and music/K-pop genres had a relatively positive effect on the four dependent variables, as these genres had a high interaction level, which led to higher viewer engagement than other genres. From the results, it was seen that the viewers who watch the genres based on conversation tend to purchase more than any other genres. Since talking with viewers is a key content of these genres, streamers encourage their viewers to interact. Psychological attachment towards streamer is important to watch a streamer's broadcasts consistently. (Hu et al., 2017). Therefore, the effect of viewer engagement on gift purchase is obvious. On the other hand, sports, news, finance, and life genres had a negative effect on the amount of gifts purchased. These genres have the characteristics of conventional broadcast content, and this characteristic can be attributed to the low interactivity level of the existing broadcasts. It leads to lower viewer engagement and gift purchase.

We looked at the impact of previous experience of purchasing items on gift purchase behaviors. As shown in the results of the experience of "Quick View," it affected gift purchase, but did not have impact on the amount of purchases. Viewers consume "Quick View" for convenience purposes, such as removal of pre-roll commercials. It is valuable for viewers, but has no direct benefits for streamers. In contrast, it assumes that "Star Balloon" and "Gold" are purchased to give gifts, which are helpful for the streamers. The purchase of "Quick View" may reduce the barrier to purchase other paid items, and thus, it becomes easy for users to buy a "Star Balloon" once after buying a "Quick View." However, since there is a lack of socialization motive in the purchase of "Quick View," it may be difficult to encourage to buy more "Star Balloons."

7. Conclusions

This study has several theoretical and practical contributions in the following aspects. Above all, it identifies viewers' behaviors through the lens of engagement and quantitatively analyzes the relationship between engagement and purchase behaviors in live video streaming platforms, which is newly emerging and replacing the existing styles of media consumption. This study overcomes the limitations of the previous studies, which did not take into account the differences of viewers' social interactions and their impact on gift purchase, which is the main revenue source for this service (excluding advertisement revenue). In addition, this research assumes viewers' channel engagement was not considered in existing media research, and develops the measurement for empirical validations. Along with content engagement, channel engagement reflects the social nature between streamers and viewers. These findings can be applicable to future research.

Through this research approach, we empirically found that social interactions or even community cultures in live video streaming services may be important to understand viewers' behaviors and increase revenues. Social interactions in live video streaming can be interpreted as community spirits, which enables a viewer to choose and to concentrate on one or a few channels to build strong relationships (Holt and Sanson, 2014). Thus, it can be seen that both chatting and gift-giving are ways of expressing and enhancing a

Table 4
Result of logistic and negative binomial regressions.

Model	1-A	1-B	2-A	2-B
Functional form	Logistic regression		Negative binomial regression	
Dependent variable	star_purchase_ox (purchase or not)		star_purchase (the amount of gifts purchased)	
rt_chat_stream	0.119***		0.0147***	
	(0.000741)		(0.00429)	
rt_recommendation_stream	0.103***		0.0508***	
	(0.000918)		(0.00452)	
rt_chat_channel		0.162***		−0.0868***
		(0.00130)		(0.00767)
rt_recommendation_channel		0.168***		0.0812***
		(0.00158)		(0.00861)
age: 20s	1.038***	1.108***	1.182***	1.232***
	(0.0129)	(0.0130)	(0.0452)	(0.0481)
age: 30s	1.188***	1.254***	1.609***	1.708***
	(0.0138)	(0.0140)	(0.0512)	(0.0537)
age: over 40s	0.894***	0.963***	1.429***	1.506***
	(0.0138)	(0.0139)	(0.0576)	(0.0607)
gender	−0.304***	−0.245***	0.109***	0.102***
	(0.00711)	(0.00716)	(0.0303)	(0.0302)
mobile_only	−0.711***	−0.671***	−0.326***	−0.307***
	(0.00575)	(0.00587)	(0.0297)	(0.0291)
join_days	−0.000213***	−0.000164***	−0.000234***	−0.000215***
	(6.58e−06)	(6.69e−06)	(3.51e−05)	(3.47e−05)
join_days_sq	3.42e−08***	2.73e−08***	3.95e−08***	3.69e−08***
	(1.32e−09)	(1.34e−09)	(7.00e−09)	(6.72e−09)
log_sec	0.155***	0.0688***	0.302***	0.156***
	(0.00188)	(0.00186)	(0.0104)	(0.0108)
ox_star_3	1.849***	1.760***	1.119***	1.066***
	(0.00622)	(0.00636)	(0.0332)	(0.0330)
ox_gold_3	0.830***	0.712***	0.911***	0.852***
	(0.0199)	(0.0204)	(0.0481)	(0.0488)
ox_quick_3	0.0841***	0.0805***	0.0872	0.139
	(0.0107)	(0.0111)	(0.0688)	(0.0718)
2. Talks	0.167***	0.159***	0.708***	0.655***
	(0.00567)	(0.00577)	(0.0306)	(0.0305)
3. Eating/Cooking	0.128***	0.110***	0.382***	0.399***
	(0.0145)	(0.0144)	(0.0683)	(0.0708)
4. Life	−0.0274	−0.0816***	−0.290***	−0.280***
	(0.0170)	(0.0171)	(0.0702)	(0.0766)
5. Mobile game	0.196***	0.110***	−0.130	−0.123
	(0.0201)	(0.0202)	(0.0762)	(0.0742)
6. Sports	−0.189***	−0.353***	−0.350***	−0.313***
	(0.0472)	(0.0467)	(0.0959)	(0.0898)
7. Music/K-pop	0.302***	0.175***	0.240***	0.263***
	(0.0337)	(0.0344)	(0.0727)	(0.0730)
8. Animation	−0.671***	−0.647***	−0.0616	0.0363
	(0.0693)	(0.0684)	(0.134)	(0.126)
9. Education	0.409***	0.232***	−0.0963	−0.0997
	(0.0488)	(0.0498)	(0.151)	(0.139)
10. News	0.346***	0.112	−0.429***	−0.348**
	(0.0569)	(0.0579)	(0.108)	(0.110)
11. TV	−0.769***	−0.807***	0.695	0.527
	(0.0895)	(0.0894)	(0.442)	(0.324)
12. Finance	0.912***	0.770***	−0.704***	−0.660***
	(0.0410)	(0.0413)	(0.181)	(0.135)
13. Beauty/Fashion	0.611***	0.406***	0.687	0.660
	(0.0975)	(0.0970)	(0.357)	(0.395)
14. Others	−0.633**	−0.790***	0.159	−0.0665
	(0.199)	(0.201)	(0.650)	(0.573)
Constant	−5.653***	−4.842***	0.937***	2.194***
	(0.0235)	(0.0231)	(0.120)	(0.122)
Observations	2,294,837	2,294,837	224,311	224,311
McFadden's adjusted R^2	0.287	0.300	0.0285	0.0291
alpha			2.715	2.701

Robust standard errors in parentheses.
** p < 0.01.
*** p < 0.001.

viewer's social status within a group oriented towards a streamer. Live video streaming is not just a channel for media distribution, but an ecosystem as social experiences are merged into the content. Therefore, service providers should be careful of managing the social features of these platforms.

Second, genres that have not primarily broadcast in traditional media, such as talks and eating/cooking, are more profitable. Meanwhile, genres that are available in conventional television, such as sports, news, finance, and life are less profitable. This is due to the differences in the interactivity levels of the genres. Thus, platform operators and streamers should think about how to encourage their viewers to get more involved in their streams by considering live interactivity as the most distinguishing characteristic of live video streaming. In addition, the game genre has the highest viewership in the percentage of viewers and has been a major revenue source, but it has relatively low profitability, when compared with talks and eating/cooking genres. Therefore, it is necessary to enhance viewer interaction and engagement during broadcasting, by making changes in the content. The platform operators may try to extend viewers' interests to alternative genres as well.

Finally, this study discovers the potential of viewers in their 30s and female viewers, each of which is about 20% of the population sampled. The highest viewers (56.1%) of live video streaming are in their 20s, while viewers in their 30s had the highest impact on sales. It is necessary to additionally manage them to get more revenue in the form of cross and up-selling, because they have needs for social interaction and high purchasing power. Furthermore, they are the generation that grew up during the exceptional success of StarCraft I, launched in 1998 (Scholz, 2011), and thus, are suitable customers for live video streaming services, which is specialized in the game genre (with the highest percentage of viewership). Therefore, there are a lot of latent customers, and the platforms should consider new marketing strategies to shift them into actual customers. Female viewers have made lesser purchases than male viewers, even though they are more likely to decide to purchase. They do not have a sense of discomfort in purchasing, and therefore, a marketing strategy that can increase their purchasing amounts is needed.

Our results also provide further guidance for practitioners. Live video streaming services, which have a higher proportion of item (gift) sales in revenue relative to other online video platforms, need to make a plan to continuously increase their profits. One possible application is to devise new contents consumption process and user experience that can increase viewer engagement. The service providers could add a new feature or function on their platform, which can naturally enhance viewers' participation level in the processes of watching broadcastings. For example, it may be an alternative to show the number of chats, the number of channels users chat, and the number of broadcasts users chat and to give rewards accordingly. In particular, there is a difference in viewer engagement depending on the genre, and thus, this strategy must be set up separately. In addition, customized strategies depending on users' gender are also necessary. Women often buy items, but the amount is relatively small. Thus, the service providers need a way to overcome them by alleviating their risk-aversion tendency. On the other hand, for men, once bought, the amount is large, but purchasing itself is not frequent. Increasing the frequency of purchases can be a suitable strategy.

This study is not without a limitation in that gift-giving behaviors could not be measured due to the limitations in data gathering. Thus, we could not confirm that gift purchases lead to gift-giving. This study should have examined how viewer engagement leads to real-time gift-giving behaviors, because live video streaming offers a higher level of interactivity than the existing media platforms. It is a limitation of this study. Nevertheless, the act of gift purchase properly functions as a proxy for this study, because it is an antecedent of gift-giving. There is a need to study gift-giving as a dependent variable in future research.

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